

Method for Drone Operations Integration Strategies in Civil Airspace Using an Integrated Flight Simulation Environment

Abstract

This project develops a real-time risk assessment framework for unmanned aircraft operations by operationalizing key elements of the JARUS Specific Operations Risk Assessment (SORA) methodology. While SORA is traditionally applied as a static pre-flight procedure, this work transforms both the Air Risk Class (ARC) and Ground Risk Class (GRC) components into dynamic, continuously updated classifiers integrated directly into a flight simulation environment. The Ground Risk Classification system uses preprocessed population-density data from the Global Human Settlement Layer (GHSL), reclassified into SORA-compliant categories and indexed through a geospatial lookup engine capable of mapping live UAV coordinates to raster cells at 2 Hz. In parallel, the Air Risk Classification system combines KML-based controlled airspace boundaries, altitude thresholds, and a point-in-polygon algorithm to determine ARC levels during flight. The framework is connected to X-Plane via the XPlaneConnect API, enabling real-time ingestion of telemetry and automated evaluation of changing risk conditions along the trajectory. BlueSky is used to model surrounding manned-aviation traffic, providing broader context for airspace interactions and potential extensions of the system. All results are logged, visualized through Google Earth, and validated through multiple case-study flight paths over the Jakarta metropolitan area. The resulting system demonstrates how SORA can be adapted from a static planning tool into an operational risk-monitoring capability, enabling continuous risk awareness, improved transparency, and a foundation for future UTM and detect-and-avoid integration.

1. Introduction

1.1 Specific Operations Risk Assessment (SORA)

The Specific Operations Risk Assessment (SORA) framework provides a structured methodology for evaluating the risks associated with unmanned aircraft operations, with particular emphasis on air risk and ground risk. In this project, SORA is adapted into a computational form that enables both risk dimensions to be evaluated

continuously during flight simulation rather than only during pre-flight planning. The air risk component is implemented through a real-time classification system that integrates controlled airspace boundaries, altitude thresholds, and aircraft position to derive ARC-b, ARC-c, or ARC-d at every simulation step. The ground risk component is similarly operationalized by mapping the aircraft's geolocation to population density data and converting it into a dynamic GRC level that reflects the density of uninvolved persons below the flight path. By automating SORA's traditionally qualitative assessments into an ongoing, data-driven process, the system provides a more responsive and transparent understanding of operational risk as flight conditions evolve.

1.2 X-Plane

X-Plane is employed in this study as the primary flight dynamics simulator due to its high-fidelity aerodynamic modelling and support for external data communication. Using the XPlaneConnect plugin, real-time telemetry, including latitude, longitude, altitude, attitude, and groundspeed can be streamed directly into custom Python modules for risk assessment. This capability allows the simulation to act as a live surrogate for an actual UAV, with the risk classification algorithms receiving continuous updates at fixed time intervals. X-Plane also provides a controlled and repeatable environment for testing risk logic under various operational scenarios, including different routes, altitudes, and environmental conditions. By integrating X-Plane with the SORA-based ARC and GRC classifiers, the system enables a full simulation-to-analysis pipeline in which aircraft behavior and changing risk profiles can be examined simultaneously.

The initial phase will be intended to construct and test a prototype of the EHang 184 using Plane Maker, a design and configuration system available as part of the X-Plane flight simulation platform. The purpose of this step is to obtain the aircraft geometry, aerodynamic configuration, and simple control system that qualitatively captures the flight behaviour of the EHang 184 eVTOL. The resulting model will be the base of the later integration into X-Plane to simulate dynamically and continue to integrate with a fixed-base flight simulator. The EHang 184 vehicle follows a symmetric X-shaped design that has four rotor arms, which carry a total of eight propellers. To make the modeling process easier, the major components of the model were divided: fuselage, rotor arms (miscellaneous wings), propeller system, and landing gear to facilitate modular design and allow alterations to be made later in the simulation process.

According to the design reference, the fuselage length is about 2.074 meters, the maximum width is 1.018 meters at the cabin section, and the maximum height is about 1.5 meters. The design has been shaped in the form of a streamlined capsule to give a resemblance to its true aerodynamic shape of the aircraft, and the closed cockpit design. To simulate the four rotor arms of the EHang 184 that are structural supports of its electric propulsion units, the Misc Wings feature of X-Plane Plane Maker was used. The arm span was calculated using the reference drawing, the total distance between the opposing rotors was 3.899 meters and the distance between the front and the rear rotor was around 3.866 meters. The chord length was specified in relation to the assumed diameter of the rotor hub mounting structure (root chord of about 0.5 ft and tip chord of about 0.2 ft). The incidence and dihedral angles were adjusted to zero degrees. EHang 184 has eight electric propellers which are formed by four coaxial pairs which occupy the ends of the rotor arms. Each arm supports two counter-rotating propellers (one upper and one lower) to provide balanced vertical thrust and eliminate net torque on the airframe. The EHang 184 features a simple system of fixed twin-landing-gear made up of two major struts located symmetrically under the fuselage, unlike the conventional plane. The purpose of these short, stiff legs is both to act as the primary structural support during landing and takeoff, and at the same time maintain clearance of the ground to the lower fuselage and the propeller assembly.

The weight and balance modeling is used to make sure that the mass distribution of the vehicle is in accordance with the given design parameters, thus making it possible to accurately simulate the performance of the vehicle, also its stability and control. Empty weight includes the main structure, propulsion system, and other permanent parts of the vehicle. The geometric design of EHang 184 is then considered to establish the center of gravity (CG) of the plane, considering the weight distribution between the fuselage, battery, passenger, and other payload. The Plane Maker makes it possible to enter the CG location in length units, starting at the nose on each of the relevant weight points. Once the empty weight and the initial center of gravity (CG) has been determined, the next thing to do is adding the payload, which is the passengers or any other cargo. The load may also include the weight of the battery that will influence the CG.

After the entire modeling process in Plane Maker (geometry, propulsion systems, weight and balance, rotor configuration) the next step is to integrate the EHang 184 model into X-Plane to be able to test the flight simulation. This is accomplished by exporting the aircraft as a .acf file and placing the resultant file into the Aircraft folder of X-Plane in order to allow the simulator to consider the model to be a choice aircraft. Once the file is

correctly stored, the VTOL configuration must be verified, including thrust vector settings, collective and cyclic behavior, as well as rotor placement and rotation directions to prevent unintended torque or yaw during hover. The control system must be calibrated to ensure the operator inputs based on joysticks or similar peripherals get a proper mapping to the control logic of the eVTOL. Throttle is used in this arrangement as collective thrust and pitch, and roll and yaw inputs are converted to differential thrust and cyclic actions on the eight coaxial rotors, no conventional aerodynamic control surfaces being in use.

1.3 Chosen Cases

SkyDrive Inc. (“SkyDrive”), a leading electric vertical takeoff and landing (eVTOL) aircraft manufacturer based in Japan, today announced the signing of a strategic partnership with PT Whitesky Aviation (“Whitesky”), one of Indonesia’s foremost helicopter operators. The collaboration aims to accelerate the adoption of eVTOL aircraft in Indonesia and explore innovative business models that can transform both urban and industrial mobility.

Under the agreement, SkyDrive and Whitesky will jointly explore the deployment of the “SKYDRIVE” (*1) eVTOL aircraft across several use cases. These include operations based out of Cengkareng Heliport, which is owned by Whitesky and strategically located next to Soekarno-Hatta International Airport, and potential applications supporting Indonesia’s vital mining sector. As part of this early-stage commitment, Whitesky has placed a pre-order for up to 30 SKYDRIVE units, demonstrating its intent to lead in the advancement of electric air mobility within Southeast Asia.

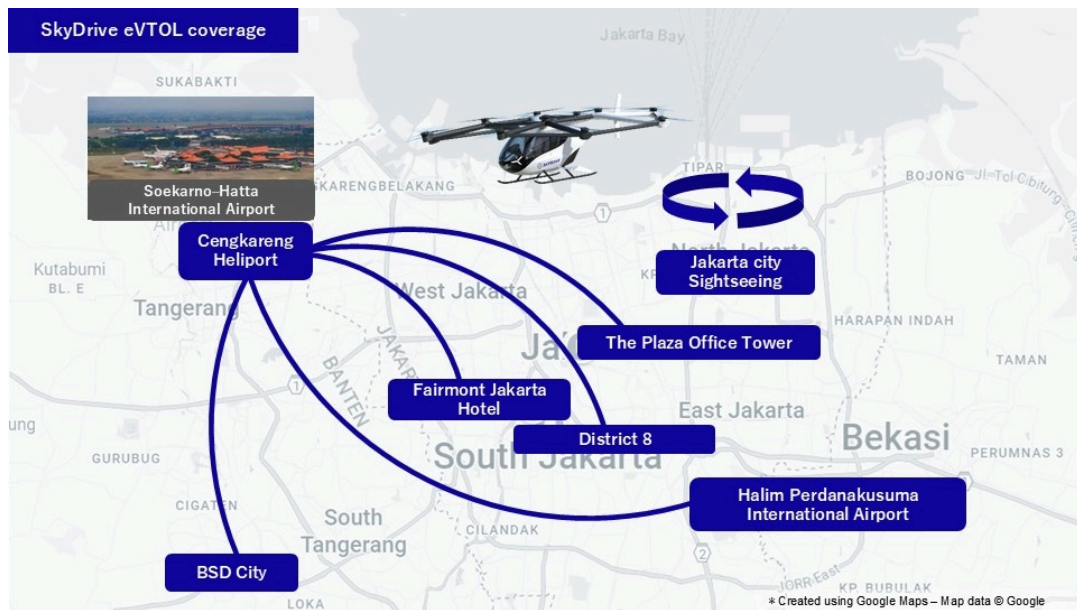


Figure 1.1 SkyDrive-Whitesky Collaboration Route Plan

The Cengkareng Heliport (Soekarno-Hatta Airport) acts as the central hub. From here, the eVTOLs are deployed to various "spokes" (destinations) across the Greater Jakarta area. Focusing on the specific route, Soekarno-Hatta (CGK) to Halim Perdanakusuma (HLP). This route is unique compared to the others because it is an Inter-Airport Transfer. It is not just about moving people to a hotel; it is about connecting two vital nodes of national transportation infrastructure.

The objective for this mission is to transition the existing helicopter shuttle service to electric vertical take-off and landing (eVTOL) aircraft, reducing noise and operational costs while bypassing Jakarta's severe ground traffic. With the distance presumably 33 km, with population densities varying from 15000 to above 50000 people/km². The route likely follows the Cawang-Pluit Toll Road or the Cengkareng-Batu Ceper-Kunciran Toll Road to minimize risk. Cutting a 2–3 hour ground trip (traffic dependent) into a ~15-20 minute flight.



Figure 1.2. SkyDrive SD-05 (eVTOL)

Table 1.1 SkyDrive SD-05 Performance

SkyDrive SD-05	
Dimensions (L x W x H)	Approx. 13 m x 13 m x 3 m
Seating Capacity	3 (1 Pilot and 2 Passengers)
Power Supply	Battery Electric
Configuration	Multicopter (12 rotors)
Kinetic Energy	Approx. ~1400 kg MTOW
Maximum Cruise Speed	~100 km/h or 27.67 m/s
Operational Flight Range	15 km

2. Methodology

2.1 Algorithm to Determine Air Risk Class (ARC)

To support real-time risk assessment based on the JARUS SORA framework, an automated real-time risk classification module based on the JARUS SORA framework was developed. The module combines **Ground Risk Classification (GRC)** and **Air Risk Classification (ARC)** to continuously evaluate the operational risk of a UAV trajectory flown in X-Plane, using population density data and controlled airspace polygons. The system is implemented in Python and runs as a loop at **2 Hz**, receiving telemetry from X-Plane via XPlaneConnect, computing GRC and ARC, logging the results, updating a dashboard, and generating a KML track for post-flight visualization.

2.1.1 Data Sources

The ARC module relies on the following inputs:

- a. **Real-time aircraft telemetry** from X-Plane 12 via XPlaneConnect, such as:
 - Latitude,
 - Longitude,
 - Altitude,
 - Heading, and
 - Groundspeed.

- b. **Controlled airspace boundaries** from official **KML files** (e.g., Halim ATZ, Soekarno-Hatta ATZ).
- c. **Ground risk category (GRC)** values generated separately by the GRC classifier, used to distinguish **urban vs rural** environments.

2.1.2 Parsing Airspace Boundaries

Airspace geometry is extracted from KML using `xml.etree.ElementTree`.

Each KML polygon is parsed into a list of longitude–latitude coordinate pairs.

This allows the system to determine whether the aircraft is currently *inside* or *outside* controlled airspace by applying a **ray-casting (point-in-polygon)** algorithm.

2.1.3 Real-Time ARC Determination

The logic follows the official **SORA Annex B Airspace Risk Model**.

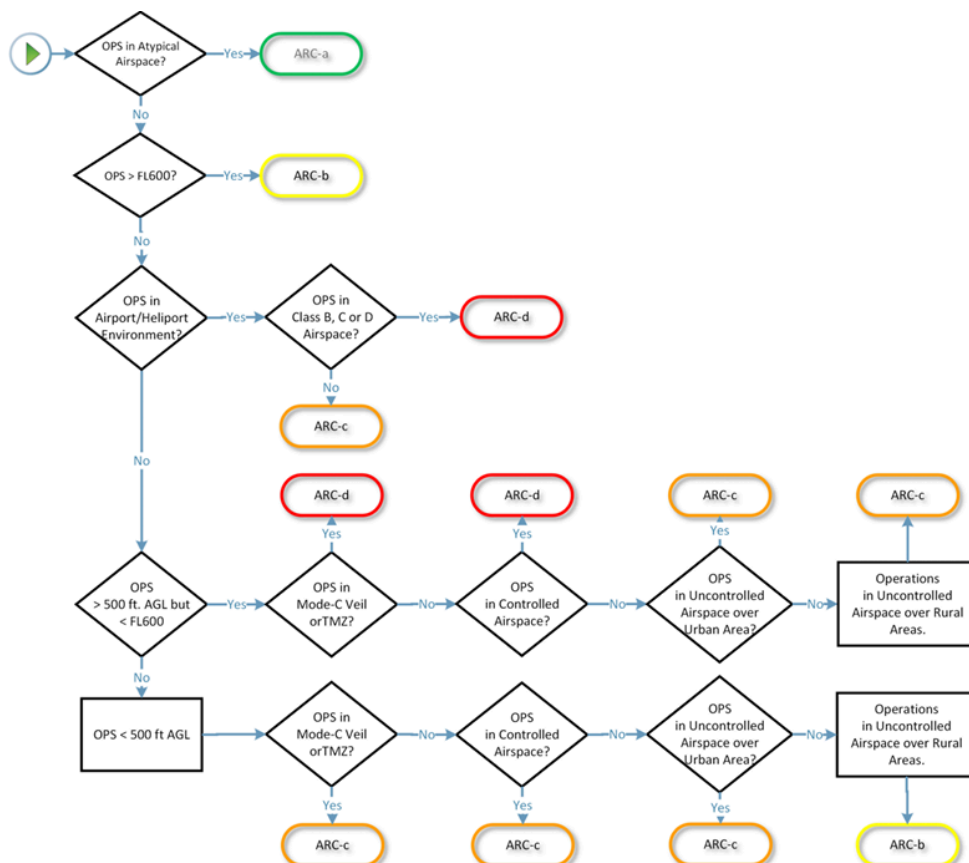


Figure 2.1 Air Risk Model from JARUS SORA 2.5

The algorithm applies a series of altitude-based branches and location checks:

1. Inside controlled airspace

- Inside Halim ATZ → classified as *Class C equivalent* → **ARC-d**
 - Inside Soekarno-Hatta ATZ → *Class A* → **ARC-c**
2. **Above FL600**
 - Operations above 60,000 ft (≈ 18 km) are automatically **ARC-b** (very low air risk).
 3. **Between 500 ft and FL600**
 - In controlled airspace → **ARC-d**
 - In uncontrolled airspace → **ARC-c**
 4. **Below 500 ft**
 - Urban / controlled → **ARC-c**
 - Rural → **ARC-b**

The ARC function returns:

- Whether the drone is in controlled airspace (`in_ctrl`)
- The ARC label (ARC-b / ARC-c / ARC-d)
- Numerical index for plotting
- A dictionary containing the reasoning rule (for transparency and logging)

This ensures every classification step is fully traceable for post-flight analysis.

2.2 Algorithm to Determine Ground Risk Class (GRC)

Ground Risk Classification (GRC) is one of the core components of the SORA framework, used to quantify the level of risk posed to uninvolved persons on the ground during a UAV operation. Traditionally, SORA defines GRC based on three primary elements: the vehicle's kinetic energy potential (characteristic dimension and maximum speed), the operational environment (population density), and any reduction factors applied through strategic mitigations.

Intrinsic UAS Ground Risk Class						
Maximum UA characteristic dimension		1m / approx. 3 ft	3 m / approx. 10 ft	8 m / approx. 25 ft	20 m / approx. 65 ft	40 m / approx. 130 ft
Maximum speed		25 m/s	35 m/s	75 m/s	120 m/s	200 m/s
Maximum iGRC population density (people/km ²)	Controlled Ground Area	1	1	2	3	3
	< 5	2	3	4	5	6
	< 50	3	4	5	6	7
	< 500	4	5	6	7	8
	< 5,000	5	6	7	8	9
	< 50,000	6	7	8	9	10
	> 50,000	7	8	Not part of SORA		

- A UA weighing less than or equal to 250 g and having a maximum speed less than or equal to 25 m/s is considered to have an iGRC of 1 regardless of population density.
- A UA expected to not penetrate a standard dwelling will get a -1 GRC reduction in Step 3 from the M1(A) sheltering mitigation when not overflying large open assemblies of people, see Annex B for additional details.

Figure 2.2 Ground Risk Determination Table from JARUS SORA 2.5

In the context of this project, the goal is to translate SORA's qualitative GRC methodology into a fully automated, real-time computational system capable of determining the intrinsic GRC (iGRC) dynamically from the UAV's simulated flight path. To accomplish this, a spatially referenced population-density map is preprocessed and integrated with Python indexing routines that convert each latitude–longitude coordinate into a discrete GRC row. This real-time GRC assignment enables a continuous assessment of ground risk as the UAV traverses regions of differing population densities.

2.2.1 Spatial Data Pre-processing (GIS Workflow)

Before integrating the population-density map with the Python classifier, a preprocessing workflow was carried out using QGIS to transform the raw global dataset into a format optimized for real-time lookup. The primary data source was the

Global Human Settlement Layer (GHSL), which provides gridded population counts at a resolution of approximately 100×100 meters.

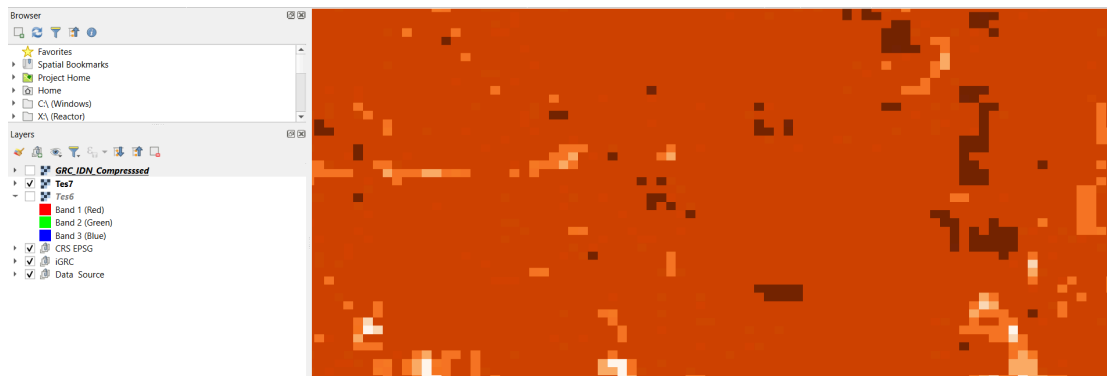


Figure 2.3 GHSL Grid Cell Data

Because global datasets are computationally heavy, the first step involved clipping the GHSL raster to the geographic boundaries of Indonesia using a national shapefile. This ensures that only relevant cells are preserved, significantly reducing the dataset size and lookup overhead.

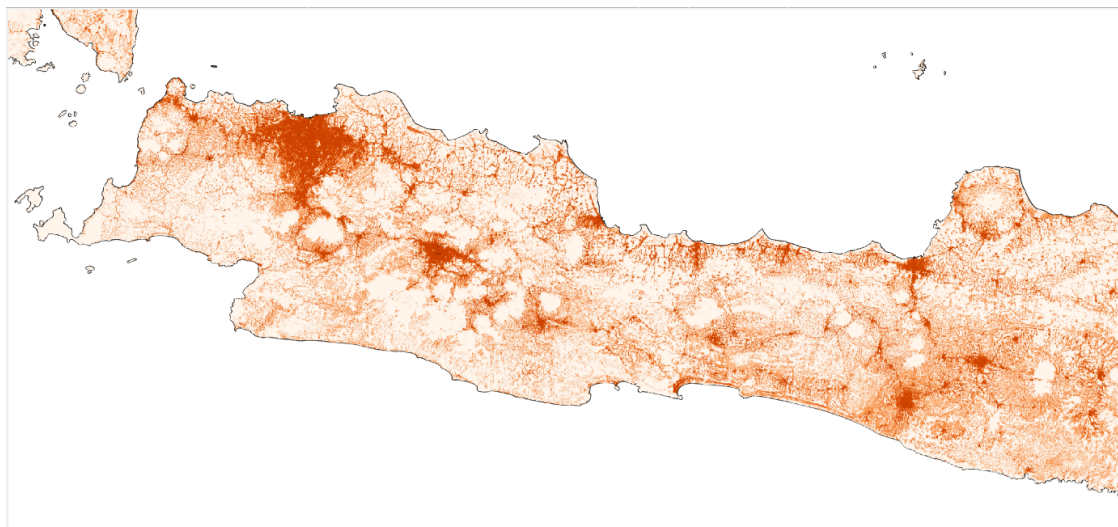


Figure 2.4 Indonesia GHSL Heat-map

Following clipping, the raster was reprojected to the World Mollweide projection (ESRI:54009) to eliminate distortion associated with latitude–longitude grids and to preserve relative population densities across regions.

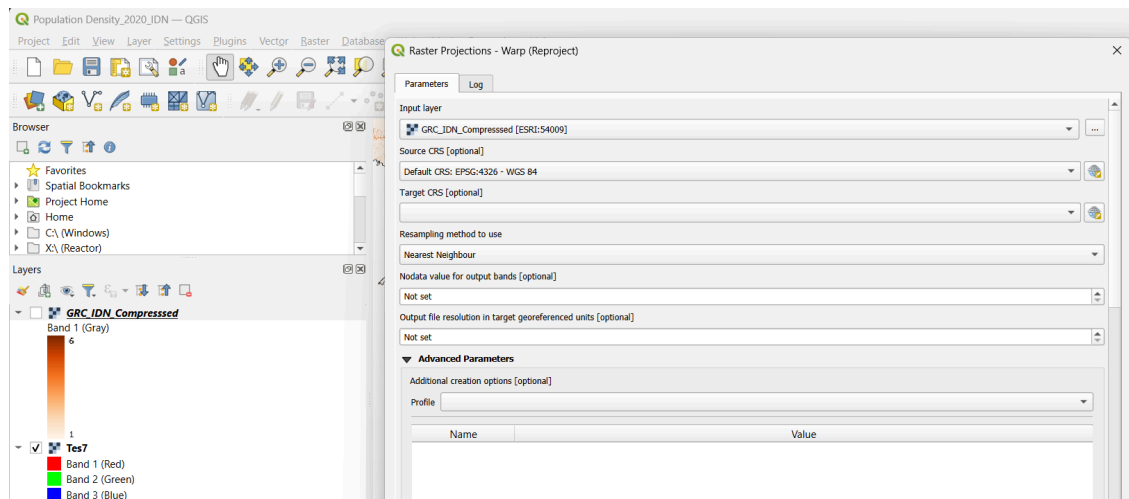


Figure 2.5 Warp Reproject

Accurate reprojection is essential, because the live risk engine later relies on coordinate transformation to map the UAV's GPS coordinates to the raster grid. Using a consistent geographic projection ensures that the density values sampled during flight accurately correspond to the underlying population dataset.

2.2.2 Raster Reclassification for SORA Integration

The GHSL raster originally contains continuous population-density values, which must be discretized into GRC-compatible categories. To achieve this, the raster was reclassified into six SORA-equivalent density bins, each representing a specific operational environment. These categories correspond to:

Table 2.1 Data Reclassification

SORA Density Category	Population Range (people/km ²)	Raster Index
Controlled Ground Area	N/A	0
SORA Row 1	< 5	1
SORA Row 2	5 – 50	2
SORA Row 3	50 – 500	3
SORA Row 4	500 – 5,000	4
SORA Row 5	5,000 – 50,000	5
SORA Row 6	> 50,000	6

This reclassification allows the real-time engine to retrieve discrete risk indices directly from the raster, eliminating the need for continuous density interpretation during flight. The resulting raster becomes a look-up table where each pixel corresponds to a SORA

GRC category. This streamlined representation is essential for ensuring that the Python module can compute risk values efficiently, especially under a 2 Hz update rate.

2.2.3 Algorithmic Integration (Python GRC Engine)

Once the raster was prepared, a dedicated Python class—referred to as the GRC engine—was developed to integrate the static GIS dataset with the dynamic UAV telemetry. During operation, the drone provides GPS coordinates in WGS84 (EPSG:4326), which must be converted into the same projection used by the raster (World Mollweide). This conversion is accomplished using the `pyproj` library, ensuring coordinate consistency.

After transformation, the projected coordinates are mapped to row–column indices within the raster matrix. This indexing process identifies the exact pixel beneath the UAV, allowing the engine to extract the corresponding SORA density index (0–6). A final mapping layer translates this raster-derived index into the intrinsic GRC row applicable for the aircraft’s fixed characteristic dimension and maximum speed. By combining these stages coordinate transformation, raster lookup, and SORA row mapping the Python module achieves a seamless and automated determination of the drone’s GRC in real time.

2.3 Live Risk Computation Module

The system includes two computers acting as the flight simulator and instructor station respectively as illustrated in Figure 2.6.

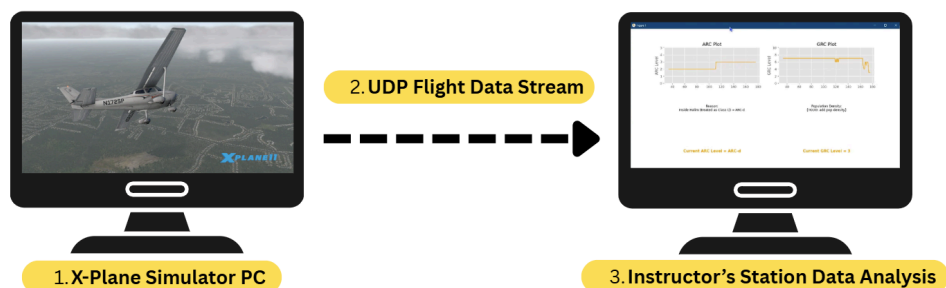


Figure 2.6 Flight simulator data stream illustration

The simulator uses a commercially available X-Plane flight simulator. Information from the X-Plane Simulator PC is transmitted to the Instructor station using the “User Datagram Protocol” or UDP. Specifically, the current research utilizes the X-Plane Connect plugin, developed by NASA, to retrieve data from X-Plane to a python

environment. X-Plane connect was originally developed to allow aircraft control and state information retrieval using numerous programming languages, including C, C++, MATLAB, or Python in real time.

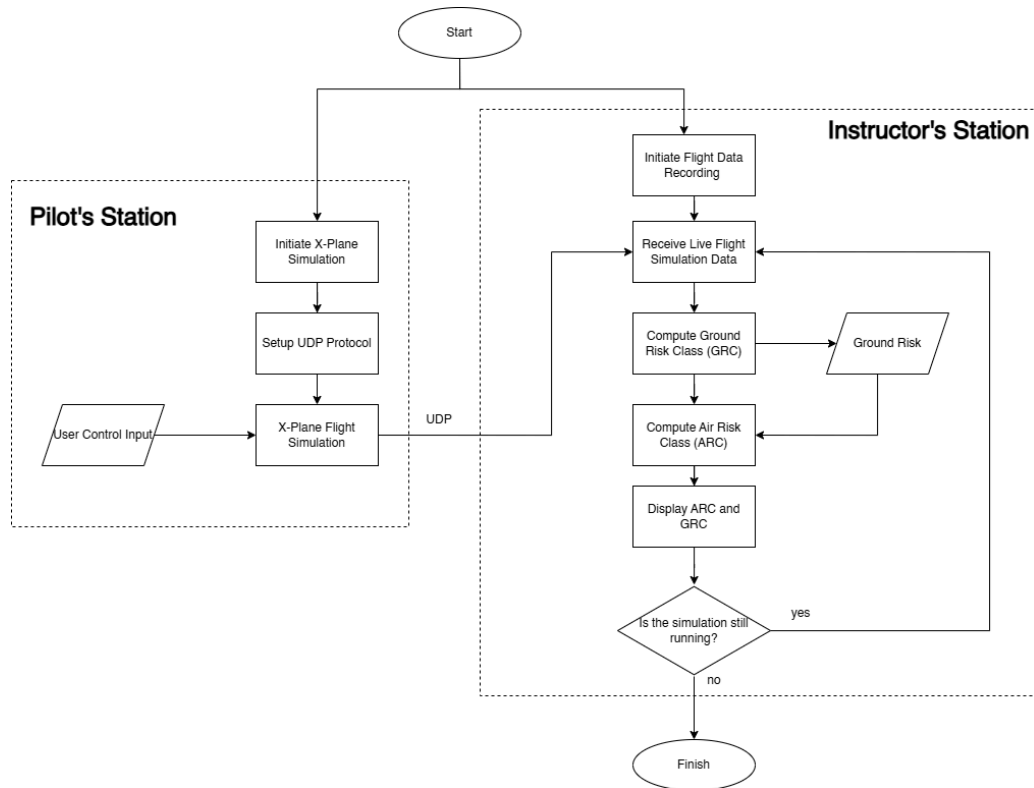


Figure 2.7 Live Risk Computation Module Flowchart

The algorithms explained in previous sections to compute ARC and GRC are integrated as illustrated in Figure 2.7. In the beginning, X-Plane Simulation is initiated in the Pilot's Station or the X-Plane PC. At the same time, the Instructor's Station initiates an empty flight data place holder. Next, a UDP connection is initiated between the two systems. As the X-Plane simulation is controlled by the pilot, flight data including latitude, longitude, and altitude information is streamed via the UDP to the instructor's station. From these data, the ARC and GRC at the current time is computed according to the previous algorithms. Firstly, GRC is computed using the population density information and current aircraft position. Next, the population density information is utilized to improve the ARC computation. Finally, the ARC, GRC, and flight information are displayed in the dashboard to the instructor.

3. Discussion and Result

3.1 Simulation Result

UAM mission between Soekarno-Hatta International Airport (CGK) and Halim Perdanakusuma Airport (HLP) in Jakarta, can be seen in the Figure 3.1.



Figure 3.1 UAM Simulation Result
North (Blue), Direct (Orange), and South (Green)
OAWKO

3.2 Air Risk Class (ARC) Analysis

Fig. 3.1 illustrates the Air Risk Class (ARC) for each route over the duration of the mission. The X-axis is the simulation time and Y-axis represents the risk level, likely mapping Level 2 to ARC-b (uncontrolled airspace/low encounter rate), Level 3 to ARC-c (controlled airspace/urban environment), and so forth.

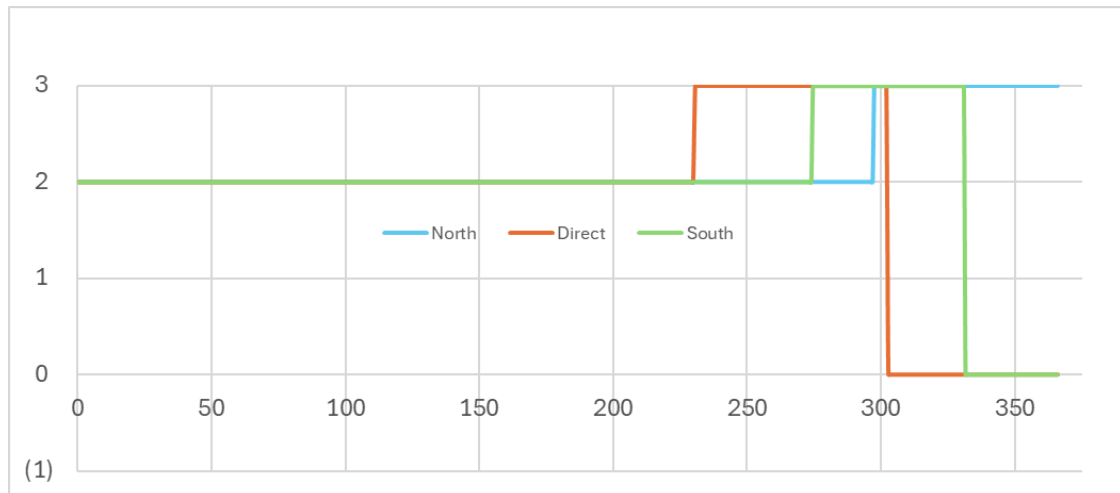


Figure 3.2 UAS Simulation ARC Result

Based on Figure 3.1, the graph shown for North route (Blue) maintains Level 2 (Low Risk) for the longest duration, only jumping to Level 3 (ARC-c) at the very end of the mission (at $t \sim 297$ s). This route avoids the controlled airspace or high-traffic zones of Halim Airport for the majority of the flight. By hugging the northern coast, it minimizes the probability of conflicting with manned aircraft traffic until the final approach. This is the safest route regarding Air Risk.

Regarding the South route (Green) it switches to Level 3 (ARC-c) earlier than the other routes ($t \sim 274$ s), but still more later than the Direct route, this means as a middle ground. It likely skirts the edge of the control zone (CTR) longer than the Direct route but eventually must enter high-traffic airspace to land. Jumping onto the Direct route (Orange) peaks to Level 3 significantly earlier than the other routes ($t \sim 230$) and remains there until the end of simulation. As a straight line, this route cuts directly through the airspace that is likely controlled or has a higher density of manned traffic. While it is the shortest path, it exposes the drone to high air collision risk for a much larger percentage of the flight time.

3.3 Ground Risk Class (GRC) Analysis

Fig 3.3 illustrates the Intrinsic Ground Risk Class (GRC), which estimates the risk to people on the ground based on population density and UAS maximum speed. A score of 7 typically indicates flight over a "Populated Environment," while scores below 4 suggest "Sparsely Populated" or "Controlled Ground" areas.

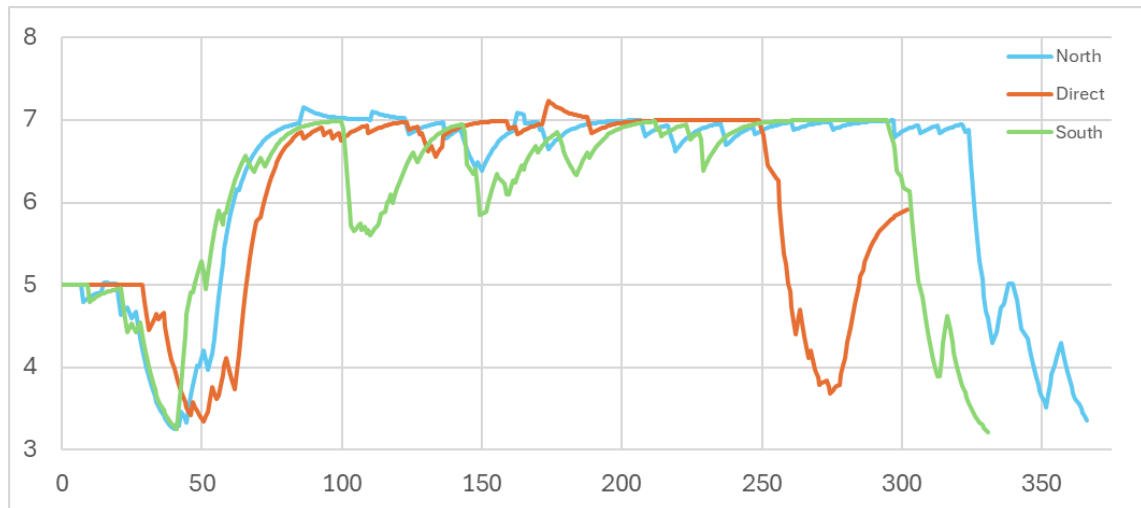


Figure 3.3 UAS Simulation GRC Result

Based on this Figure 3.3, the North route (Blue) presents the highest prolonged exposure to ground risk. After an initial takeoff climb, the risk score spikes to nearly 7 and sustains this high level for almost the entire cruise phase (from $t = 70$ to $t = 320$). This suggests that, contrary to the expectation of a safer coastal path, the route actually overflies densely populated coastal settlements or industrial areas with high human presence.

In contrast, the Direct Route (Orange) and South Route (Green) offer distinct risk profiles. The Direct Route initially spikes to ~ 7 but features a significant safety advantage during the final approach, with a massive drop in risk to ~ 3 (between $t = 250$ and $t = 290$), likely indicating flight over a low-density zone such as a green belt or water body. Furthermore, due to its shorter duration ending at $t = 300$, it results in the lowest total time exposure. Meanwhile, the South Route shows a high risk (~ 7) similar to the North but with notable fluctuations, dipping to ~ 5 around $t = 100$ - 150 . These fluctuations suggest the route passes over mixed environments like suburbs and open fields, making it slightly safer than the North route but less efficient than the Direct option.

4. Conclusion

In conclusion, the selection of the optimal flight path requires balancing the opposing safety objectives of minimizing mid-air collision probability versus reducing ground casualty risk, the overall result can be seen in Table 4.1.

Table 4.1 Overall Risk Simulation Result

Route	Air Risk (ARC)	Ground Risk (GRC)	Efficiency (Distance)
North (Blue)	Lowest (Best)	High (Sustained)	Medium
Direct (Orange)	High (Worst)	Lowest (Shortest Exposure)	High (Best)
South (Green)	Medium	Low (Fluctuate)	Medium

Trade-off

Recommendation

The Direct Route (Orange) is preferable because it minimizes the flight time and total ground risk exposure. If the UAS lacks technology, the North Route (Low) is safer to prevent mid-air collisions, provided the ground risk is mitigated (e.g., via parachute).